

Mining Social Network Graphs

Introduction

The increasing popularity of social networks is clearly demonstrated by the huge number of users acquired in a short amount of time. Social networks now have gathered hundreds of millions of users, prominent among them being Facebook, MySpace, Twitter, LinkedIn, Instagram, etc. Further data from these networks increases in volumes at every instance in time. Facebook has over one billion active users with thousands of new users added every day. The number of interactions added everyday amounts to millions. Moreover, most of these networks allow interactions using different media like voice, images, and even videos. Thus, social network data is “Big Data”.

Massive quantities of data on very large social networks are now freely available from many social networks like blogs, social networking sites, newsgroups, chat rooms, etc. These networks typically contain substantial quantities of information at the level of the individual node. Mining these structures for patterns yields knowledge that can effectively be used for predictions and decision-making.

One very fundamental task needed for social network mining is community detection which basically identifies dense sub-graphs from the social network graphs.

Applications of Social Network Mining

Applications of social network mining encompass numerous fields. A few popular ones are enumerated here:

1. Viral marketing is an application of social network mining that explores how individuals can influence the buying behavior of others. Viral marketing aims to optimize the positive word-of-mouth effect among customers. Social network mining can identify strong communities and influential nodes. It can choose to spend more money marketing to an individual if that person has many social connections.
2. Similarly in the e-commerce domain, the grouping together of customers with similar buying profiles enables more personalized recommendation engines. Community discovery in mobile ad-hoc networks can enable efficient message routing and posting.
3. Social network analysis is used extensively in a wide range of applications, which include data aggregation and mining, network propagation modeling, network modeling and sampling, user attribute and behavior analysis, community-maintained resource support, location-based interaction analysis, social sharing and filtering, recommender systems, etc.
4. Many businesses use social network analysis to support activities such as customer interaction and analysis, information system development analysis, targeted marketing, etc.

Social Networks as a Graph

When the objective is to mine a social network for patterns, a natural way to represent a social network is by a graph. The graph is typically very large, with nodes corresponding to the objects and the edges corresponding to the links representing relationships or interactions between objects. One node indicates one person or a group of persons.

As an example, consider a set of individuals on a social networking site like LinkedIn. LinkedIn allows users to create profiles and “connections” with each other in an online social network which may represent real-world professional relationships. Thus, finding a person on a LinkedIn network is like finding a path in a graph representing the social network. Similarly, a company may want to find the most influential individual in a social network to perform targeted advertising. The hope is that more influential a person, more people can be influenced to buy their goods. This means identifying nodes with very high out-degree in the graph representing the social network.

Finding communities or clustering social networks, all lead to the standard graph algorithms when the social network is modeled as a graph. This section will give an overview of how a social network can be modeled as a graph. A small discussion of the different types of graphs is also provided.

➤ *Social Networks*

The web-based dictionary Wikipedia defines a social network as “A social structure made of nodes that are generally individuals or organizations”. A social network represents relationships and flows between people, groups, organizations, computers or other information/knowledge processing entities. The term “Social Network” was coined in 1954 by J. A. Barnes. Examples of social networks include Facebook, LinkedIn, Twitter, Reddit, etc.

The following are the typical characteristics of any social network:

1. In a social network scenario, the nodes are typically people. But there could be other entities like companies, documents, computers, etc.
2. A social network can be considered as a **heterogeneous** and **multi-relational** dataset represented by a graph. Both nodes and edges can have attributes. Objects may have class labels.
3. There is at least one relationship between entities of the network. For example, social networks like Facebook connect entities through a relationship called friends. In LinkedIn, one relationship is “endorse” where people can endorse other people for their skills.
4. In many social networks, this relationship need not be yes or no (binary) but can have a degree. That means in LinkedIn, we can have a degree of endorsement of a skill like say novice, expert, etc. Degree can also be a real number.
5. We assume that social networks exhibit the property of non-randomness, often called locality. Locality is the property of social networks that says nodes and edges of the graph tend to cluster in communities. This condition is the hardest to formalize, but the intuition is that the relationships tend to cluster. That is, if entity A is related to both B and C, then there is a higher probability than average that B and C are related. The idea is most relationships in the real

worldstend to cluster around a small set of individuals.

➤ **Social Network Graphs**

A natural representation of social network data is to view its structure as a graph. The basic process is as follows. Entities are converted to nodes in the graph. The nodes in this graph can be of different types, for example, people, organizations and events. This model, thus, slightly deviates from the standard model of a graph in which the node set is one type. Edges in the graph correspond to the relations between entities. If there is a degree associated with the relationship, this degree is represented by labeling the edges. Edges can be one-directional, bi-directional and are not required to be binary.

Example 1

Figure 1 shows a small graph of the “followers” network of Twitter. The relationship between the edges is the “follows” relationship. Jack follows Kris and Pete shown by the direction of the edges. Jack and Mary follow each other shown by the bi-directional edges. Bob and Tim follow each other as do Bob and Kris, Eve and Tim. Pete follows Eve and Bob, Mary follows Pete and Bob follows Alex. Notice that the edges are not labeled, thus follows is a binary connection. Either a person follows somebody or does not.

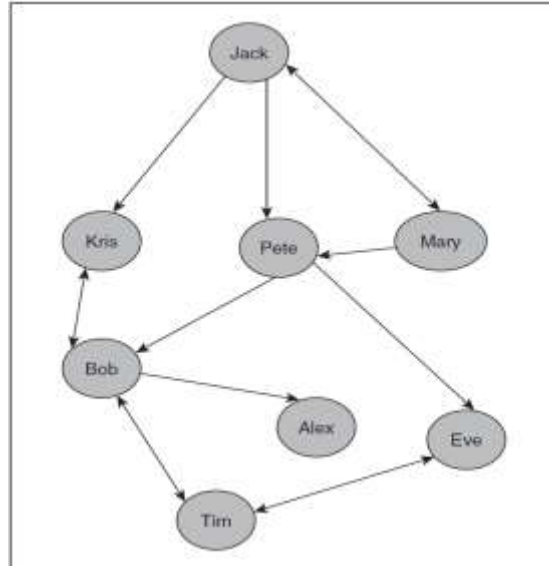


Figure 1 A social graph depicting follows relationship in Twitter.

Example 2

Consider LiveJournal which is a free on-line blogging community where users declare friendship to each other. LiveJournal also allows users to form a group which other members can then join. The graph depicted in Fig. 2 shows a portion of such a graph. Notice that the edges are undirected, indicating that the “friendship” relation is commutative.

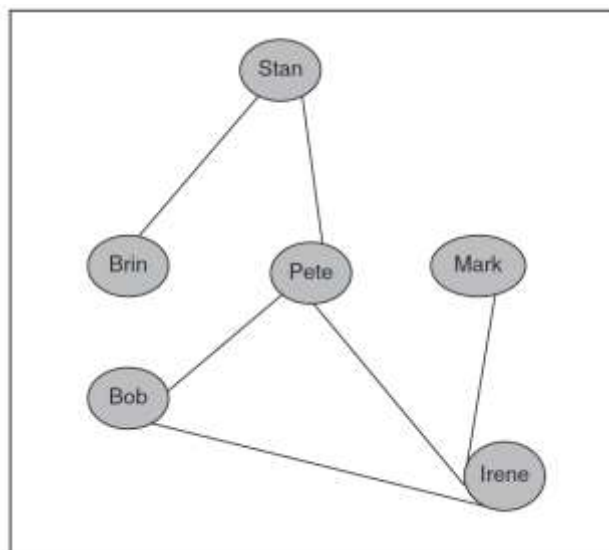


Figure 2 Social graph of LiveJournal.

Example 3

As a third example, we consider DBLP. Computer science bibliography provides a comprehensive list of research papers in computer science. As depicted in Fig. 3, the graph shows a co-authorship relationship where two authors is connected if they publish at least one paper together. The edge is labeled by the number of papers these authors have co-authored together.

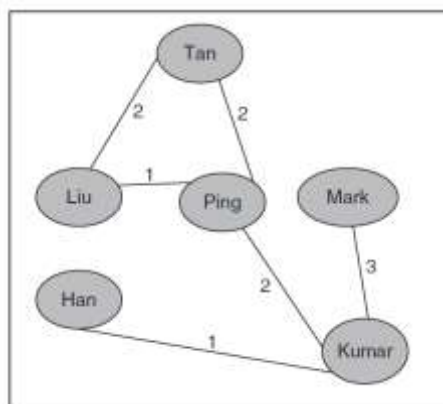


Figure 3 Social graph of DBLP.

Types of Social Networks

While Facebook, Twitter and LinkedIn might be the first examples that come to mind when discussing social networks, these examples in no way represent the full scope of social networks that exist. In this section, we give a brief overview of different scenarios that may result in social networks. This is by no means an exhaustive list. The interested user can refer to “<http://snap.stanford.edu/data>” for a rather large list of social network examples.

- 1. Collaboration Graph:** Collaboration graphs display interactions which can be termed collaborations in a specific setting; co-authorships among scientists and co-appearance in movies by actors and actresses are two examples of collaboration graphs. Other examples include the Wikipedia collaboration graph (connecting two Wikipedia editors if they have ever edited the same article), or in sport the “NBA graph” whose vertices are players where two players are joined by an edge if they have ever played together on the same team.
- 2. Who-Talks-to-Whom Graphs:** These model a type of social network also called “Communication” network. The best example is the Microsoft Instant Messenger (IM) graph which captures a snapshot of a large group of users engaged in several billion conversations in a time period. The edges between the two nodes of the graph, say, A and B denote that A “talked-to” B in this time period. Edges can be labeled with the strength of the communication too. Examples of similar networks include the email communication networks with edges representing communication between the entities. One good example is the Enron email communication network which captured all the email communication within a dataset of around half million emails. Nodes of the network are email addresses and if an address i sent at least one email to address j , the graph contains an undirected edge from i to j . Similar networks have also been constructed from records of phone calls: Researchers have studied the structure of *call graphs* in which each node is a phone number, and there is an edge between two if they engaged in a phone call over a given observation period.
- 3. Information Linkage Graphs:** Snapshots of the web are central examples of such network datasets. Nodes are Web pages and directed edges represent links from one page to another. The web consists of hundreds of millions of personal pages on social networking and blogging sites, each connecting to web pages of interest to form communities. One good example is the Berkeley-Stanford web graph where nodes represent pages from berkeley.edu and stanford.edu domains and directed edges represent hyperlinks (possible connections) between them.

Other examples include product co-purchasing networks (nodes represent products and edges link commonly co-purchased products), Internet networks (nodes represent computers and edges communication), road networks (nodes represent intersections and edges roads connecting the intersections), location-based online social networks, Wikipedia, online communities like Reddit and Flickr, and many such more. The list is endless.
- 4. Heterogeneous Social Networks:** Finally, a social network may have heterogeneous nodes and links. This means that the network may consist of different types of entities (multi-mode network)

and relationships (multi-relational network). For example, a simple co-purchasing network may have product nodes and customer nodes. Edges may exist if a customer purchases a product. Similarly, in a typical e-commerce platform like Amazon, it is reasonable to assume the existence of distributor information to add a third type of node. We now have edges between a customer and a product, a product sold by a distributor, and also a customer and a distributor. Different customers could buy the same product from different distributors. The UCI KDD Archive has a social network dataset about movies which contains eight different node types, including actor, movie, role, studio, distributor, genre, award and country and links between them. The natural way to represent such information is as a k -partite graph for some $k > 1$, where k is the number of different node types. In general, a k -partite graph consists of k disjoint sets of nodes, with no edges between nodes of the same set. One special graph is a bi-partite graph where $k = 2$.

Example 4

Figure 4 shows an example of a tri-partite graph where $k = 3$. We can assume three types of nodes to indicate actors, movies and directors. Notice no edge exists between the nodes of the same type.

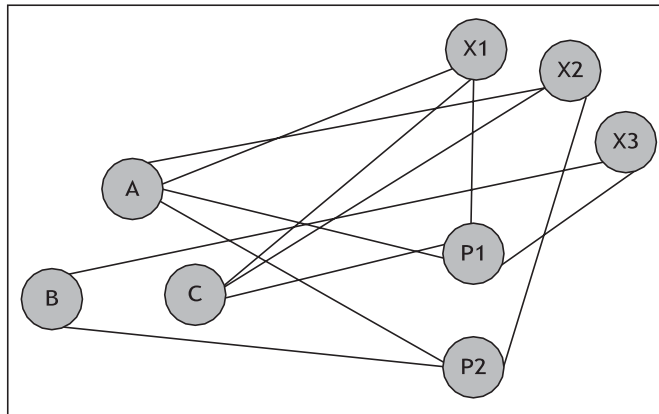


Figure 4 A tri-partite social graph.

Clustering of Social Graphs

The discovery of “communities” in a social network is one fundamental requirement in several social network applications. Fundamentally, communities allow us to discover groups of interacting objects (i.e., nodes) and the relations between them. Typically in a social network we can define a “community” as a collection of individuals with dense relationship patterns within a group and sparse links outside the group.

There are many reasons to seek tightly knit communities in networks; for instance, target marketing schemes can benefit by identifying clusters of shoppers and targeting a campaign wholly customized for them. Communities allow us to discover groups of interacting objects (i.e., nodes) and the relations between them. For example, in co-authorship networks, communities correspond to a set of common scientific disciplines and a closely knit set of researchers on that topic. Identifying clusters of customers with similar interests in the network of purchase relationships between customers and products of online retailers enables to set up efficient recommendation systems.

Community can be considered as a summary of the whole network, thus easy to visualize and understand. Sometimes, community can reveal the properties without releasing the individual privacy information.

One simple method for finding communities in a social graph is to use clustering techniques on the social graph. In this section, we first indicate why traditional clustering techniques like K -means and Hierarchical clustering cannot just be extended to social graphs. We shall then discuss a popular graph clustering technique called the Girvan–Newman Algorithm.

➤ *Applying Standard Clustering Techniques*

As we are aware, to cluster similar data points in a social graph into groups, we need to decide on some measure of similarity or dissimilarity. When the social graph has labeled edges, we can use that value itself as a measure of similarity or dissimilarity between the two nodes based on what the edge represents. For example, if the social graph depicts the DBLP network discussed earlier, the edge indicates the number of papers the authors have co-authored and thus use that as some notion of similarity between the two authors. But most social graphs would be unlabeled and thus we cannot design a similarity measure.

One solution will be to have a binary distance measure, 1 for no edge and 0 for edge existing between two nodes. Any two values could be used as long as the distance for no edge is greater than the distance for an edge. The main problem is that such a measure will not satisfy the triangular inequality property. If edge AB and edge BC exist and no edge AC exists then $\text{dist}(AB) + \text{dist}(BC) < \text{dist}(AC)$. This is because the notion of distance does not have an inherent meaning in a social network. We can still try to solve this problem by choosing appropriate distance values for existence and non-existence of an edge. One example will be to use a distance of 1 for an edge and distance of 1.5 for a missing edge. We shall now use traditional clustering methods with this distance measure.

Example 5

Consider a sample social graph as depicted in Fig. 5. Visually we can see that $\{1, 2, 3, 4\}$ and $\{5, 6, 7, 8\}$ form two clusters (communities). But it can equally be argued that using the above mentioned distance measure, $\{5, 6, 7\}$, $\{6, 7, 8\}$, $\{4, 5, 6\}$, $\{1, 2, 3\}$ is an equally good clustering. This is an overlapping clustering which would never be identified by a traditional hierarchical clustering algorithm.

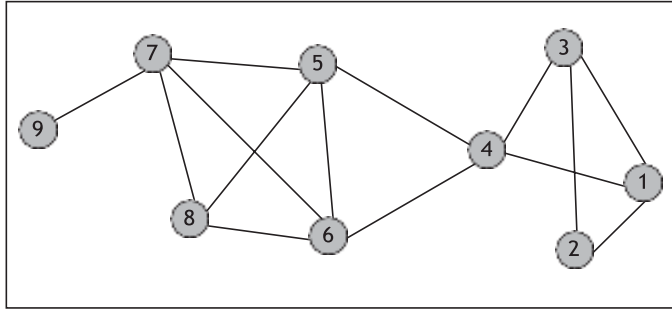


Figure 5 Sample social graph.

Further hierarchical clustering with the above-discussed distance measure has a tendency to separate single peripheral vertices from the communities to which they should rightly belong. For example, node 9 in Fig. 5. If a node is connected to the rest of a network by only a single edge, then instead of adding to the cluster formed by $\{5, 6, 7, 8\}$, it is left isolated. Further, the presence of a node like node 5 in Fig. 5 allows the hierarchical clustering algorithm to combine all the nodes $\{1, 2, 3, 4, 5, 6, 7, 8\}$ into one large cluster.

Since the distance measure is same for all edges, these problems will be magnified in a large graph leading to meaningless clusters. Let us now consider how a partition-based algorithm like *K*-means will work on clustering social networks. We shall see that here also, the fact that all edges are at the same distance will introduce random results which will lead to wrong assignment of nodes to a cluster.

Example 6

Consider again the graph of Fig. 5. *K*-means clustering will output clusters based on the initial choice of centroids. Let us say $K = 2$, and initial choices for centroids are, say, 1 and 3 which belong to the same cluster. Because every edge has same cost, we will land with wrong clusters. (Convince yourself that this is true.) Assume the initial centroids are 4 and 8. Since both nodes 5 and 6 are equidistant from both the centroids, they could be clustered with either even though they should be clustered with 8.

The above problem is compounded in a large social network where any choice of *K*-means, even far off in the graph, will lead to a bad clustering because at some point a large number of nodes will be equidistant from many means. Further like hierarchical clustering, *K*-means does not discover overlapping clusters.

➤ Introducing “Betweenness” Measure for Graph Clustering

A major problem with using traditional clustering techniques was in the choice of distance measure used. These distance measures did not capture the essence of what a cluster in a social graph means. Our aim is to extract dense sub-graphs from the social graph. It will be the case that in a graph we can have a set of dense sub-graphs connected to other dense sub-graphs through a minimum set of edges. If we can identify and remove these connecting edges, we land up with the dense sub-graphs that form our communities.

This motivates the definition of a distance measure which is called “Edge Betweenness” associated with each edge in the social graph. Edge betweenness (EB) is defined in such a way that the edges that are likely to connect different dense regions (communities) of the graph have higher betweenness scores than edges within the communities. Hence, by identifying and progressively discarding edges with high betweenness scores, one can extract communities from a social network.

A formal definition of “edge betweenness” can be given as follows: *For an edge e in a graph, edge betweenness of e is defined as the number of shortest paths between all node pairs (v_i, v_j) in the graph such that the shortest path between v_i and v_j passes through e .*

If there are k different shortest paths between v_i and v_j we divide the number by “ k ”.

Example 7

Consider the graph of Fig. 5 reproduced again as Fig. 6.

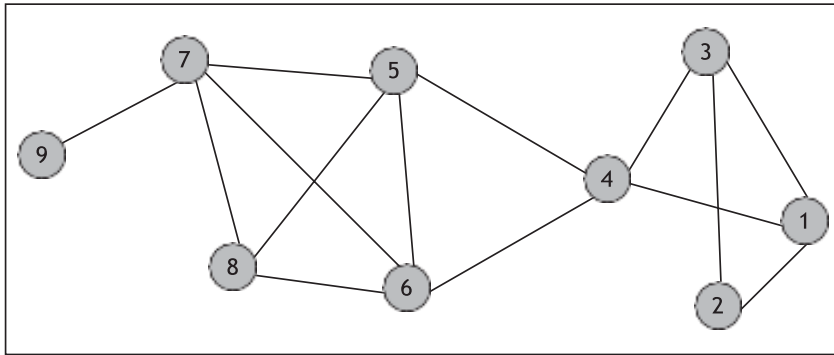


Figure 6 A social graph.

We shall denote edge betweenness of edge e as $EB(e)$. We have

1. $EB(1, 2)$ is 4 ($= 6/2 + 1$), as all the shortest paths from 2 to $\{4, 5, 6, 7, 8, 9\}$ go through either $e(1, 2)$ or $e(2, 3)$, and $e(1, 2)$ is the shortest path between 1 and 2.

2. $EB(4,5) = \{\text{from } 9, 7, 8 \text{ to } 4, 3, 2, 1 \text{ either through } e(4,5) \text{ or } e(4, 6) = 12/2 = 6; \text{ plus from } 5 \text{ to } 1, 2, 3, 4 = 4. \text{ This yields } 6 + 4 = 10.\}$
3. Similarly $EB(4,6) = 10$; $EB(5,7) = EB(6, 7) = 6$ and so on.

As an exercise try and compute the EB of all other edges of the graph.

➤ *Girvan–Newman Algorithm*

Girvan and Newman proposed a hierarchical divisive clustering technique for social graphs that use EB as the distance measure. The basic intuition behind this algorithm is that edges with EB are the most “vital” edges for connecting different dense regions of the network, and by removing these edges we can naturally discover dense communities. The algorithm is as follows:

2. Calculate EB score for all edges in the graph. We can store it in a distance matrix as usual.
3. Identify the edge with the highest EB score and remove it from the graph. If there are several edges with the same high EB score, all of them can be removed in one step. If this step causes the graph to separate into disconnected sub-graphs, these form the first-level communities.
4. Re-compute the EB score for all the remaining edges.
5. Repeat from step 2. Continue until the graph is partitioned into as many communities as desired or the highest EB score is below a pre-defined threshold value.

For the graph of Fig. 6, the following will be the order of edge removal (see Fig. 7):

- $e(4, 5)$ (or $e(4, 6)$). By removing $e(4, 5)$.
- We re-compute the EB. $e(4, 6)$ has the highest betweenness value: 20. This is because all shortest paths between $\{1,2,3,4\}$ and $\{5,6,7,8,9\}$ pass $e(4, 6)$.
- Continuing in this way we get three communities as $\{1, 2, 3, 4\}$, $\{5, 6, 7, 8\}$, $\{9\}$.

The major cost for the above algorithm is finding the EB score for every edge in the graph. We need to find the number of shortest paths from all nodes. Girvan and Newman proposed a faster algorithm based on use of Breadth First Search (BFS) algorithm. An overview of their algorithm is given below. The reader is referred to any graph theory book for more details.

For each node N in the graph

1. Perform breadth-first search of graph starting at node N .
2. Determine the number of shortest paths from N to every other node.
3. Based on these numbers, determine the amount of flow from N to all other nodes that use each edge.
4. Divide sum of flow of all edges by 2.

The references describe the method in greater detail.

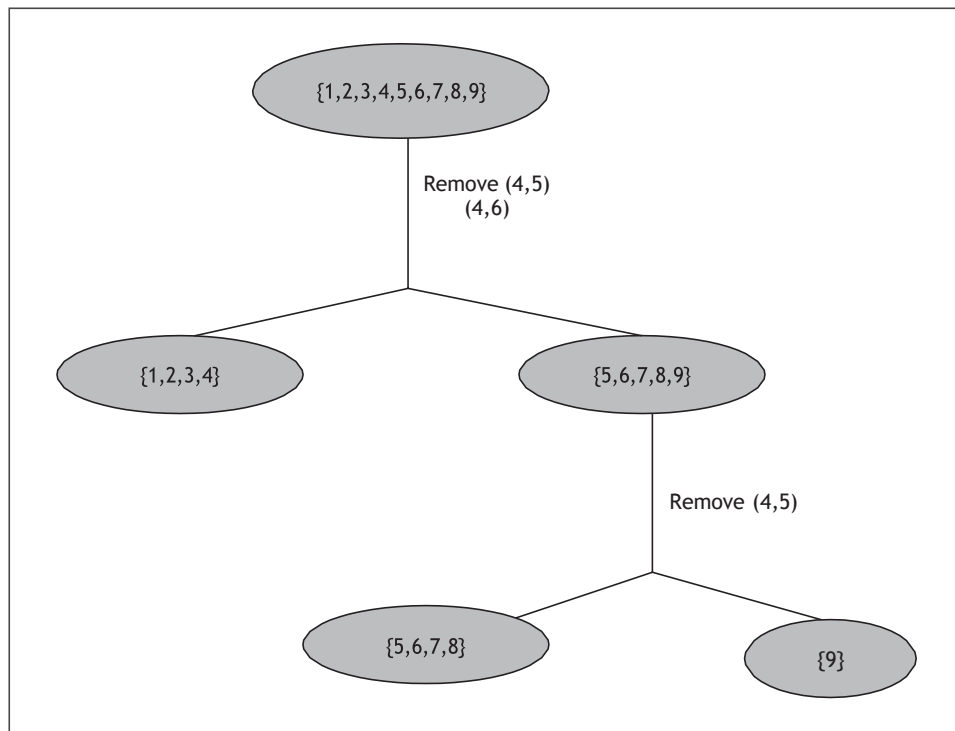


Figure 7 Dendrogram illustrating Girvan–Newman Algorithm.